



Kalman Delta Networks 2: Associative Memory with Damped Rotational Dynamics

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Source Code: <https://github.com/ngocbh/kalman-delta-networks>

Abstract

Linear attention compresses a sequence into a fixed-size associative memory. Existing delta-rule mixers learn how strongly that memory should decay, but their positive real transitions primarily control *survival*: they do not provide a process mechanism for moving surviving information between latent directions. We introduce *Kalman Delta Networks 2* (KDN-2), which equips uncertainty-aware associative memory with damped rotational dynamics. Each fixed two-dimensional mode has a radius that controls lifetime and an angle that controls phase. Learned measurement maps determine which modes are observed and corrected, while read maps determine visibility. When the uncertainty-shaped write and read phases align appropriately, a write can occupy a latent quadrature with little immediate readout and later become observable, without conflating persistence with accessibility. Its deterministic rotational transport is a moving-coordinate representation of the accumulated rotary addressing used by Mamba-3; KDN-2 differs by coupling this shared geometry to an innovation-driven, covariance-shaped correction. We formulate KDN-2 as a linear-Gaussian state-space model and derive its exact tokenwise Kalman residual update, dense covariance propagation, rotational composition law, and survival-visibility decomposition. Exact dense filtering remains computationally demanding. As an implementation detail, we therefore propose processing KDN-2 in chunks using a co-rotating frame, freezing dense-prior gains within each chunk, preserving causal sequential residuals through a WY form, and committing the corresponding Joseph error covariance at the boundary. This design preserves deterministic rotational transport exactly while approximating within-chunk covariance feedback and process-noise timing. We state these boundaries explicitly and give falsifiable tests for phase-aware recall, cue-controlled observability, interference, approximation error, and complete systems cost.

1. Introduction

Self-attention is a flexible content-addressing mechanism: every query can compare itself with all preceding keys and retrieve their values [Vaswani et al., 2017]. This flexibility comes from retaining the token history. The autoregressive key-value cache grows with context length, and processing a full sequence requires quadratic query-key interactions. Linear attention instead compresses the prefix into a fixed-size recurrent state [Katharopoulos et al., 2020]. Its scaling is attractive, but compression turns attention into an online memory-management problem. Once many observations share one state, the model must decide what survives, what is overwritten, and what a query can currently observe.

DeltaNet treats the recurrent state as an associative map. It reads the value stored at the current key and writes only the residual [Schlag et al., 2021; Yang et al., 2024b]. Gated DeltaNet and KDA additionally learn positive real transitions that decay the memory globally or by channel [Yang et al., 2024a; Kimi Team, 2025]. These mechanisms create a bank of retention timescales. They answer *how*

much of a memory remains, but they do not provide a process mechanism for moving the surviving state between latent directions. A slow channel remains in the same coordinate; it is visible whenever the query overlaps that coordinate. A fast channel vanishes and cannot later be restored by the process alone.

This observation suggests that memory dynamics have three distinct roles. *Survival* asks whether a write retains appreciable norm. *Phase* asks where that surviving write moves in latent space. *Visibility* asks whether the current read direction observes it. Positive diagonal decay mainly controls the first quantity. Changing queries can still change visibility, but the process itself cannot rotate an association into a latent quadrature, preserve it there, and move it toward a readable phase. Conflating these roles makes it difficult to express active/archive behavior, delayed visibility, or phase-selective retrieval.

Kalman associative memory provides the right framework for separating them. It regards the recurrent matrix as an estimate of a latent, drifting key-value map. Each token is a noisy linear observation, the process model predicts how the map moves, and a covariance records which address directions are supported by evidence. The Kalman gain then produces an uncertainty-weighted delta write: uncertain directions adapt quickly, while well-supported directions resist noisy corrections. Importantly, the process transition is not merely a forget gate. It is the model’s explicit hypothesis about how stored associations move between observations.

We introduce *Kalman Delta Networks 2* (KDN-2), whose process model consists of damped two-dimensional rotations. Each mode has complex-conjugate poles $\rho e^{\pm i\theta}$ represented in real arithmetic. The radius ρ controls survival, the angle θ controls phase motion, and learned measurement and read maps control which modes are written, corrected, and observed. A cross-quadrature write can therefore have small immediate readout while retaining norm and becoming visible after later rotations. A bank of modes yields damped oscillatory responses rather than only zero-frequency exponential decay.

Damped rotational transport itself is not the distinction from prior sequence models. Mamba-3 accumulates phase in its key and query maps and thereby realizes the same open-loop relative-phase kernel [Lahoti et al., 2026]. Section 4.4 gives the exact change of coordinates between that representation and KDN-2’s incremental state rotation. The models separate at the memory update: Mamba-3 uses additive exponential-trapezoidal writes, whereas KDN-2 uses the innovation and an evolving covariance to determine what is corrected and by how much.

KDN-2 remains a Kalman delta network, not an unrelated spectral state-space model. The exact tokenwise recursion predicts both the associative memory and its dense covariance through the rotational process, forms the uncertainty-optimal residual write, and reads the corrected state. Dense uncertainty matters because a measurement immediately creates anisotropy and correlations between modes. Those correlations can redirect later gains, so the observation address does not by itself determine the write phase. The transition, covariance, measurement map, and read map must be analyzed together.

The rotational construction also has sharp limitations. Rotation changes temporal coding and interference; it does not create additional capacity. A fixed oscillator schedules visibility by elapsed time but does not by itself retrieve an arbitrary item on a semantic cue. Query-controlled observability remains necessary, and later Kalman corrections can erase a nominally hidden quadrature. We therefore distinguish the open-loop process response from the actual closed-loop lineage after intervening measurements, and design evaluations that randomize cue times and recall order.

The exact tokenwise recurrence defines KDN-2’s reference semantics, but its dense covariance follows a fine-grained Riccati recurrence. Our proposed implementation design addresses this systems obstacle without redefining the model. Within each chunk, we transform keys and queries into a co-rotating frame, compute dense-prior gains together, and preserve causal sequential residuals with a WY form. At the boundary, a Joseph update records the ensemble error covariance of the local frozen-gain estimator under a delayed-noise surrogate. Deterministic rotational transport remains token-timed; the proposed design approximates covariance feedback and the timing of process noise. Chunking is thus an implementation detail and an explicit approximation, not the motivation or definition of KDN-2.

Our contributions are:

- We formalize survival, phase, and visibility as distinct properties of recurrent associative memory and derive their roles in open-loop and closed-loop retrieval.
- We introduce KDN-2, an exact tokenwise Kalman associative-memory model with damped rotational transitions, phase-aware measurement and read maps, and dense uncertainty-aware residual writes.
- We prove that its rotational transport is coordinate-equivalent to Mamba-3’s accumulated rotary addressing, then isolate the different update rules. We also characterize hidden and observable quadratures and distinguish scheduled resurfacing from recall on cue.
- We derive a causal chunkwise implementation in a co-rotating frame. It retains dense uncertainty at boundaries and exact deterministic rotational transport while making its within-chunk filtering approximations explicit.

The suffix in KDN-2 denotes this richer process geometry for Kalman Delta Networks. It is unrelated to the architectural suffix in Gated DeltaNet-2 (GDN-2; [Hatamizadeh et al., 2026](#)).

2. Preliminaries

2.1. Linear Associative Memory and Delta Updates

Let $S_t \in \mathbb{R}^{d \times m}$ be a recurrent associative memory with address width d and value width m . A key–value pair (k_t, v_t) writes to the memory and a query q_t reads

$$o_t = S_t^\top q_t. \quad (1)$$

Additive linear attention uses $S_t = S_{t-1} + k_t v_t^\top$, which cannot explicitly remove a conflicting association. DeltaNet instead takes a residual-gradient step on the instantaneous recall error,

$$S_t = S_{t-1} + \beta_t k_t (v_t - S_{t-1}^\top k_t)^\top. \quad (2)$$

Gated variants first transform the state,

$$\hat{S}_t = D_t S_{t-1}, \quad S_t = \hat{S}_t + \beta_t k_t (v_t - \hat{S}_t^\top k_t)^\top, \quad (3)$$

where $D_t = \alpha_t I$ in Gated DeltaNet and $D_t = \text{diag}(\alpha_t)$ in KDA. These transitions provide multiple survival timescales, but the scalar gain β_t does not depend on accumulated uncertainty.

2.2. Associative Memory as a Filtering Problem

Let S_t^* be a latent associative map. A linear–Gaussian model writes

$$\begin{aligned} S_t^* &= D_t S_{t-1}^* + W_t, & W_t &\sim \mathcal{N}(\mathbf{0}, \Omega_t), \\ v_t &= S_t^{*\top} k_t + e_t, & e_t &\sim \mathcal{N}(\mathbf{0}, r_t I). \end{aligned} \quad (4)$$

The matrix-normal notation is columnwise: value coordinates are conditionally independent and share one key-space covariance. Let $\mathcal{F}_t$ denote the observations available through token t . The minimum conditional mean-squared-error estimate is

$$S_t = \arg \min_{\tilde{S}} \mathbb{E} \left[\|S_t^* - \tilde{S}\|_F^2 \mid \mathcal{F}_t \right] = \mathbb{E}[S_t^* \mid \mathcal{F}_t]. \quad (5)$$

Given posterior mean S_{t-1} and covariance P_{t-1} , the exact Kalman reference is

$$\begin{aligned} \hat{S}_t &= D_t S_{t-1}, & \hat{P}_t &= D_t P_{t-1} D_t^\top + \Omega_t, \\ \kappa_t &= \frac{\hat{P}_t k_t}{r_t + k_t^\top \hat{P}_t k_t}, & S_t &= \hat{S}_t + \kappa_t (v_t - \hat{S}_t^\top k_t)^\top, \\ P_t &= (I - \kappa_t k_t^\top) \hat{P}_t, & o_t &= S_t^\top q_t. \end{aligned} \quad (6)$$

The gain in (6) is obtained by minimizing the conditional estimation-error objective, equivalently the trace of the candidate error covariance. Thus the Kalman update is a delta rule whose write direction is preconditioned by the uncertainty accumulated from the entire preceding stream. Appendix A derives the gain directly from this objective.

2.3. From KDN to KDN-2

The predecessor KDN construction interprets the delta rule as a Kalman correction and makes its uncertainty state scan-compatible through scalar or diagonal structure. KDN-2 keeps that uncertainty-aware interpretation but changes the process geometry: its latent associative map moves through damped rotational modes, so persistence and phase are no longer the same quantity. The exact tokenwise filter below defines the model. Section 5 later derives an approximate implementation; that implementation is not a second KDN-2 variant.

Mixer	Write geometry	Process geometry	Primary role
DeltaNet / GDN / KDA	learned scalar residual gain	identity, scalar decay, or positive diagonal decay	overwrite and retention
KDN	scalar- or diagonal-uncertainty gain	positive real scalar or diagonal dynamics	uncertainty-aware retention
KDN-2	Kalman gain in a lifted latent address space	damped rotations with complex-conjugate poles	survival, phase, and visibility

3. Beyond Geometric Forgetting

3.1. Process-Only Survival and Visibility

Consider a residual written at time τ through

$$U_\tau = \kappa_\tau \delta_\tau^\top, \quad \delta_\tau = v_\tau - \hat{S}_\tau^\top h_\tau. \quad (7)$$

Here h_τ is the measurement address; it may be the external key or a learned lift of that key. To isolate the process model, first suppress all later measurement corrections. The contribution propagated to $t > \tau$ is

$$\tilde{S}_{t \leftarrow \tau}^{\text{proc}} = \Phi_{t:\tau} \kappa_\tau \delta_\tau^\top, \quad \Phi_{t:\tau} = D_t D_{t-1} \cdots D_{\tau+1}. \quad (8)$$

For a read address g_t , its process-only output is

$$\tilde{o}_{t \leftarrow \tau}^{\text{proc}} = \delta_\tau \underbrace{\kappa_\tau^\top \Phi_{t:\tau}^\top}_{\text{visibility at time } t} g_t. \quad (9)$$

Equations (8)–(9) separate two questions that a scalar retention score conflates: $\|\Phi_{t:\tau}\kappa_\tau\|_2$ measures how much of the write survives, while its alignment with $\mathbf{g}_t$ measures how much is currently visible.

For positive diagonal dynamics,

$$\Phi_{t:\tau} = \text{diag}\left(\prod_{s=\tau+1}^t \alpha_{s,1}, \dots, \prod_{s=\tau+1}^t \alpha_{s,d}\right), \quad 0 \leq \alpha_{s,i} \leq 1. \quad (10)$$

Every scalar mode is nonincreasing in magnitude. Different coordinates provide different timescales, and a changing query can still alter their visibility, but the process cannot move surviving state between coordinates or generate oscillatory phase. A fixed change of basis does not change this fact: a transition with positive real eigenvalues still contains only zero-frequency exponential modes.

3.2. The Closed Loop

Process-only survival is necessary but not sufficient for retrieval. Expanding a Kalman residual update gives

$$\mathbf{S}_t = \underbrace{(\mathbf{I} - \kappa_t \mathbf{h}_t^\top) \mathbf{D}_t}_{\mathbf{A}_t} \mathbf{S}_{t-1} + \kappa_t \mathbf{v}_t^\top. \quad (11)$$

Let

$$\Psi_{t:\tau} = \mathbf{A}_t \mathbf{A}_{t-1} \cdots \mathbf{A}_{\tau+1}. \quad (12)$$

Holding subsequent inputs fixed, the actual contribution of write τ to a later output is

$$\mathbf{o}_{t \leftarrow \tau} = \delta_\tau \kappa_\tau^\top \Psi_{t:\tau}^\top \mathbf{g}_t. \quad (13)$$

The difference between $\Phi_{t:\tau}$ and $\Psi_{t:\tau}$ is essential. A process mode can retain norm yet be erased or mixed by later observations. We therefore use (9) to identify what the transition makes possible and (13) to evaluate what the complete memory actually retrieves.

4. Kalman Delta Networks 2

KDN-2 enriches the latent process geometry while retaining the ordinary linear–Gaussian filtering rule. Its defining model is tokenwise: the process controls survival and phase, while distinct measurement and read maps control which latent directions are corrected and observed.

4.1. Latent Measurement and Read Geometry

Let $\mathbf{S}_t^* \in \mathbb{R}^{d_s \times d_v}$ be the latent associative map. We distinguish external keys and queries from their latent-state addresses,

$$\mathbf{h}_t = \mathbf{E}_t \mathbf{k}_t, \quad \mathbf{g}_t = \mathbf{F}_t \mathbf{q}_t, \quad \mathbf{E}_t, \mathbf{F}_t \in \mathbb{R}^{d_s \times d_k}. \quad (14)$$

The measurement address $\mathbf{h}_t$ determines which directions are observed and corrected. The read address $\mathbf{g}_t$ determines which directions enter the output. Their separation makes being hidden from the current read a precise property rather than a synonym for forgetting.

For each value coordinate, KDN-2 assumes

$$\begin{aligned} \mathbf{S}_t^* &= \mathbf{D}_t \mathbf{S}_{t-1}^* + \mathbf{W}_t, & \text{Cov}(\mathbf{W}_{t,:j}) &= \mathbf{\Omega}_t, \\ \mathbf{v}_t &= \mathbf{S}_t^{*\top} \mathbf{h}_t + \boldsymbol{\xi}_t, & \boldsymbol{\xi}_t &\sim \mathcal{N}(\mathbf{0}, r_t \mathbf{I}_{d_v}). \end{aligned} \quad (15)$$

The noises are zero mean and mutually independent. The token-local quantities $\mathbf{D}_t$, $\mathbf{\Omega}_t$, $\mathbf{E}_t$, $\mathbf{F}_t$, and r_t are known when conditioning on the current input.

4.2. Exact Tokenwise Recursion

Given the posterior mean and shared key-space covariance from token $t - 1$, KDN-2 predicts

$$\hat{\mathbf{S}}_t = \mathbf{D}_t \mathbf{S}_{t-1}, \quad \hat{\mathbf{P}}_t = \mathbf{D}_t \mathbf{P}_{t-1} \mathbf{D}_t^\top + \mathbf{\Omega}_t. \quad (16)$$

With innovation $\delta_t = \mathbf{v}_t - \hat{\mathbf{S}}_t^\top \mathbf{h}_t$, the correction and read are

$$\boxed{\begin{aligned} \kappa_t &= \frac{\hat{\mathbf{P}}_t \mathbf{h}_t}{r_t + \mathbf{h}_t^\top \hat{\mathbf{P}}_t \mathbf{h}_t}, \\ \mathbf{S}_t &= \hat{\mathbf{S}}_t + \kappa_t \delta_t^\top, \\ \mathbf{P}_t &= (\mathbf{I} - \kappa_t \mathbf{h}_t^\top) \hat{\mathbf{P}}_t (\mathbf{I} - \kappa_t \mathbf{h}_t^\top)^\top + r_t \kappa_t \kappa_t^\top, \\ \mathbf{o}_t &= \mathbf{S}_t^\top \mathbf{g}_t. \end{aligned}} \quad (17)$$

Under (15), $\mathbf{S}_t = \mathbb{E}[\mathbf{S}_t^* \mid \mathcal{F}_t]$. The gain is derived from the conditional mean-squared-error objective, not assumed in advance: Appendix A shows that it is the unique minimizer of that objective, equivalently of the trace of the post-correction error covariance. Equation (17) is the reference semantics of KDN-2.

Dense covariance matters even though the process is structured. A single observation creates anisotropy and cross-coordinate correlations; the next write direction is $\hat{\mathbf{P}}_t \mathbf{h}_t$, not generally $\mathbf{h}_t$. Projecting after each observation to a scalar or diagonal family changes both uncertainty and the phase of later writes.

4.3. Damped Rotational Process

Define

$$\mathbf{R}(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}. \quad (18)$$

Partition $d_s = \sum_{a=1}^M 2r_a$ into fixed mode groups and let

$$\mathbf{D}_t = \text{blockdiag}_{a=1}^M [\rho_{t,a} \mathbf{R}(\theta_{t,a}) \otimes \mathbf{I}_{r_a}], \quad 0 < \rho_{t,a} \leq 1. \quad (19)$$

A stable token-local parameterization is

$$\rho_{t,a} = \exp(-\lambda_a \Delta_{t,a}), \quad \theta_{t,a} = \omega_a \Delta_{t,a} + \gamma_{t,a}, \quad \lambda_a \geq 0, \quad \Delta_{t,a} \geq 0, \quad (20)$$

where ω_a is a base frequency and $\gamma_{t,a}$ is an optional bounded phase increment. The initial design uses fixed learned frequencies and sets $\gamma_{t,a} = 0$; dynamic phase increments are an ablation because one kick moves every association sharing that mode. A block-isotropic process noise,

$$\mathbf{\Omega}_t = \text{blockdiag}_{a=1}^M [\nu_{t,a} \mathbf{I}_{2r_a}], \quad \nu_{t,a} \geq 0, \quad (21)$$

makes the additive noise update cheap. The fixed-plane structure of $\mathbf{D}_t$ is what lets a dense covariance prediction be evaluated in $O(d_s^2)$ rather than by a generic cubic similarity transform.

Proposition 4.1: Damped Rotational Composition

For mode a , define $\Phi_{t:\tau}^{(a)} = D_t^{(a)} \cdots D_{\tau+1}^{(a)}$. Then

$$\Phi_{t:\tau}^{(a)} = \left(\prod_{s=\tau+1}^t \rho_{s,a} \right) R \left(\sum_{s=\tau+1}^t \theta_{s,a} \right) \otimes I_{r_a}, \quad (22)$$

and, for every x in that mode,

$$\|\Phi_{t:\tau}^{(a)} x\|_2 = \left(\prod_{s=\tau+1}^t \rho_{s,a} \right) \|x\|_2. \quad (23)$$

Thus the radii determine survival independently of accumulated phase; visibility additionally depends on the read direction.

Proof. See Appendix D.

For constant ρ_a and θ_a , the eigenvalues are $\rho_a e^{\pm i\theta_a}$ and the amplitude half-life is

$$L_{1/2,a} = \frac{\log 2}{-\log \rho_a} \quad (24)$$

when $\rho_a < 1$. If b_a and c_a are the actual two-dimensional slices of a write and a later read, the process response at lag ℓ is

$$\begin{aligned} h_a(\ell) &= c_a^\top [\rho_a R(\theta_a)]^\ell b_a \\ &= \rho_a^\ell [A_a \cos(\ell\theta_a) + B_a \sin(\ell\theta_a)]. \end{aligned} \quad (25)$$

A bank of modes therefore provides damped oscillatory responses, rather than only a bank of zero-frequency exponential decays. Rotation changes temporal coding and interference; it does not increase the number of independent values that a fixed-size memory can store.

4.4. Relation to Mamba-3: Equivalent Rotational Coordinates, Different Updates

KDN-2 and Mamba-3 place phase in apparently different parts of their recurrences. KDN-2 transports the associative state with incremental rotations, whereas Mamba-3 applies accumulated rotary maps to its write and read addresses [Lahoti et al., 2026]. The two parameterizations are exactly related by a time-dependent orthogonal change of coordinates.

To make the relation explicit, consider one Mamba-3 SISO head and transpose its recurrent state into the address-by-value convention used here. After absorbing address normalization and biases into the keys and queries, and omitting only the direct skip and output gate, its memory recurrence can be written

$$\begin{aligned} A_t &= \alpha_t A_{t-1} + \beta_t \tilde{k}_{t-1} v_{t-1}^\top + \gamma_t \tilde{k}_t v_t^\top, & o_t^{\text{mem}} &= A_t^\top \tilde{q}_t, \\ \tilde{k}_t &= R_t k_t, & \tilde{q}_t &= R_t q_t, \end{aligned} \quad (26)$$

where R_t is the block-orthogonal map formed from accumulated phases. Mamba-3’s learned exponential–trapezoidal discretization uses

$$\alpha_t = \exp(a_t \Delta_t), \quad \beta_t = \alpha_t \Delta_t (1 - \eta_t), \quad \gamma_t = \Delta_t \eta_t, \quad a_t < 0, \quad 0 \leq \eta_t \leq 1. \quad (27)$$

Define a moving-frame state and the relative rotation

$$S_t^{\text{M3}} = R_t^\top A_t, \quad U_t = R_t^\top R_{t-1}. \quad (28)$$

Substituting (28) into (26) gives, without approximation,

$$\boxed{\begin{aligned} S_t^{\text{M3}} &= \underbrace{\alpha_t U_t}_{D_t^{\text{M3}}} \left[S_{t-1}^{\text{M3}} + \Delta_t (1 - \eta_t) k_{t-1} v_{t-1}^\top \right] + \Delta_t \eta_t k_t v_t^\top, \\ o_t^{\text{mem}} &= (S_t^{\text{M3}})^\top q_t. \end{aligned}} \quad (29)$$

If the planes of R_t have accumulated angles $\phi_{t,a}$, then

$$U_t = \text{blockdiag}_a [R(\phi_{t-1,a} - \phi_{t,a})]. \quad (30)$$

Thus $D_t^{\text{M3}} = \alpha_t U_t$ is exactly a damped rotational transition of the form in (19); the sign of the angle only records whether phase is assigned to the addresses or the state.

The same identity appears directly in the open-loop kernel. For $\Phi_{t:\tau}^{\text{M3}} = D_t^{\text{M3}} \cdots D_{\tau+1}^{\text{M3}}$,

$$\begin{aligned} \Phi_{t:\tau}^{\text{M3}} &= \left(\prod_{s=\tau+1}^t \alpha_s \right) R_t^\top R_\tau, \\ k_\tau^\top (\Phi_{t:\tau}^{\text{M3}})^\top q_t &= \left(\prod_{s=\tau+1}^t \alpha_s \right) (R_\tau k_\tau)^\top (R_t q_t). \end{aligned} \quad (31)$$

Accumulated rotary addressing and incremental rotational state transport therefore produce the same damped relative-phase interaction. The two trapezoidal endpoint writes change the causal weight assigned to a token, but not this coordinate identity.

Where the equivalence ends. This equivalence concerns deterministic transport, not the complete memory update. Expanding the KDN-2 mean correction gives

$$S_t = (I - \kappa_t h_t^\top) D_t S_{t-1} + \kappa_t v_t^\top, \quad \kappa_t = \frac{\hat{P}_t h_t}{r_t + h_t^\top \hat{P}_t h_t}. \quad (32)$$

Mamba-3 instead uses additive trapezoidal outer-product writes; it has no innovation subtraction, Riccati covariance, or covariance-shaped gain. Its MIMO variant replaces a SISO write with a rank- R sum but remains additive. Mamba-3 also uses a scalar radius per SISO head in this recurrence, while KDN-2 permits mode-specific radii and process uncertainty. Hence the models share an exactly equivalent rotational geometry, but not an equivalent correction rule. KDN-2's distinction is coupling that geometry to uncertainty-shaped residual correction and closed-loop associative memory.

4.5. Hidden Quadratures and Cue-Controlled Recall

Let $e_{\text{vis}} = (1, 0)^\top$ and $e_{\text{hid}} = (0, 1)^\top$. A cross-quadrature diagnostic has

$$b = e_{\text{hid}}, \quad c = e_{\text{vis}}, \quad h_{\text{delay}}(\ell) = -\rho^\ell \sin(\ell\theta). \quad (33)$$

It is invisible at insertion and becomes visible as the plane rotates. For the actual filter, however, the write is $\kappa_t \delta_t^\top$, so immediate invisibility requires $g_t^\top \kappa_t = 0$. Because $\kappa_t \propto \hat{P}_t h_t$, a measurement address does not prescribe the write phase independently of uncertainty.

The two-step observability matrix for one plane read through e_{vis} is

$$\begin{bmatrix} e_{\text{vis}}^\top \\ e_{\text{vis}}^\top \rho R(\theta) \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \rho \cos \theta & -\rho \sin \theta \end{bmatrix}. \quad (34)$$

It has full rank when $\rho \sin \theta \neq 0$: a direction may be hidden now but recoverable later. This is a process/readout statement. The closed-loop product in (12) can still destroy that direction.

A fixed oscillator supplies *scheduled resurfacing*, not arbitrary recall on a semantic cue. KDN-2 therefore makes query-controlled observability primary: F_t emits both quadratures and a causal gate over modes, allowing an ordinary token to suppress a persistent mode and a recall cue to expose it. Token-dependent phase increments can complement this mechanism, but they cannot provide content selection because a phase kick acts on all associations in a mode. Randomized cue times and recall order are consequently necessary tests.

4.6. Layer Parameterization

The surrounding layer follows the same controlled shell as its delta-rule baselines: learned query, key, and value projections, the same short convolution, output gate, gated normalization, and output projection. Latent measurement and read lifts implement (14); the read lift includes one sigmoid gate per rotational mode. Observation noise is positive and token local,

$$r_t = r_{\min} + \text{softplus}(w_r^\top x_t + b_r), \quad r_{\min} > 0. \quad (35)$$

We initialize $S_0 = \mathbf{0}$ and $P_0 = \mu^{-1} \mathbf{I}$ for learned $\mu > 0$. The main control uses the identical paired state, lifts, gates, and parameter budget with every $\theta_a = 0$. Fixed nonzero frequencies are initialized from a range of target delays, while their radii are initialized with lifetimes longer than those delays. Signed-real poles, dynamic phase increments, and alternative covariance structures are explicit ablations, not additional names for the model.

5. Chunkwise Implementation of KDN-2

The model in Section 4 is defined at token time. A direct implementation, however, needs P_t before it can form $P_t h_{t+1}$, creating a fine-grained dense Riccati dependency. We obtain matrix-oriented work by expressing deterministic within-chunk dynamics in a common co-rotating frame and freezing only the uncertainty used to form local gains. This is a proposed approximate implementation of KDN-2, not a second model. The local frame below is the chunk-restricted version of the moving-coordinate transformation in Section 4.4.

5.1. Co-Rotating Chunk Frame

Let chunk c begin with (S_c^-, P_c^-) and contain $C_c \leq C$ tokens. The structured transition $D_{c,t}$ is applied before token t . Define

$$\Phi_{c,0} = \mathbf{I}, \quad \Phi_{c,t} = D_{c,t} D_{c,t-1} \cdots D_{c,1}. \quad (36)$$

Each prefix is represented by cumulative log-radii and angles, so no generic dense product is formed. Ignoring within-chunk process noise for the moment, write the physical latent map as $S_{c,t}^* = \Phi_{c,t} X_c^*$ and pull addresses back to the entry frame:

$$\tilde{h}_{c,t} = \Phi_{c,t}^\top h_{c,t}, \quad \tilde{g}_{c,t} = \Phi_{c,t}^\top g_{c,t}. \quad (37)$$

Then

$$S_{c,t}^{*\top} h_{c,t} = X_c^{*\top} \tilde{h}_{c,t}, \quad S_{c,t}^{*\top} g_{c,t} = X_c^{*\top} \tilde{g}_{c,t}. \quad (38)$$

Thus the proposed algorithm preserves deterministic token-timed damping and rotation. It applies structured prefix transforms and the final forward map, never $\Phi_{c,t}^{-1}$.

Stack the transported addresses as

$$\tilde{H}_c = [\tilde{h}_{c,1}, \dots, \tilde{h}_{c,C_c}], \quad \tilde{Q}_c = [\tilde{g}_{c,1}, \dots, \tilde{g}_{c,C_c}], \quad (39)$$

and let row t of V_c be $v_{c,t}^\top$. In this frame the proposed algorithm uses the local observation model

$$V_c = \tilde{H}_c^\top X_c^\star + \Xi_c, \quad \text{Cov}(\Xi_{c,:j}) = R_c = \text{diag}(r_{c,1}, \dots, r_{c,C_c}). \quad (40)$$

5.2. Frozen Dense Gains and Causal Residuals

All raw gains use the same dense entry covariance:

$$\begin{aligned} U_c &= P_c^- \tilde{H}_c, & \Gamma_c &= R_c + \tilde{H}_c^\top U_c, \\ \Lambda_c &= \text{Diag}(\text{diag}(\Gamma_c)), & L_c &= U_c \Lambda_c^{-1}. \end{aligned} \quad (41)$$

Column t is

$$\ell_{c,t} = \frac{P_c^- \tilde{h}_{c,t}}{r_{c,t} + \tilde{h}_{c,t}^\top P_c^- \tilde{h}_{c,t}}. \quad (42)$$

It is the scalar Kalman gain against the chunk-entry prior in the co-rotating frame. The approximation is that earlier observations do not contract the covariance used for later raw gains.

The residuals remain sequential. Define

$$\Delta_c = V_c - \tilde{H}_c^\top S_c^-, \quad (43)$$

and

$$M_c = \text{tril}(\tilde{H}_c^\top L_c - 1), \quad T_c = I + M_c, \quad \bar{A}_c = T_c^{-1}. \quad (44)$$

The unit-lower WY solve and the committed entry-frame mean are

$$Z_c = \bar{A}_c \Delta_c, \quad \bar{S}_c^+ = S_c^- + L_c Z_c. \quad (45)$$

This is algebraically identical to applying the frozen columns of L_c with causal sequential residuals. The read-after-write outputs are

$$O_c = \tilde{Q}_c^\top S_c^- + \text{tril}(\tilde{Q}_c^\top L_c) Z_c, \quad (46)$$

where tril includes the diagonal. No later key, value, query, or transition enters an earlier output.

5.3. Boundary Error and Published State

The effective gain is

$$L_{c,\text{eff}} = L_c \bar{A}_c, \quad \bar{S}_c^+ = S_c^- + L_{c,\text{eff}} \Delta_c. \quad (47)$$

Let

$$A_c = I - L_{c,\text{eff}} \tilde{H}_c^\top. \quad (48)$$

Under the delayed-noise local model in (40), the Joseph ensemble-MSE covariance of the frozen-gain estimator is

$$\boxed{\bar{P}_c^+ = A_c P_c^- A_c^\top + L_{c,\text{eff}} R_c L_{c,\text{eff}}^\top} \quad (49)$$

Equivalently,

$$\bar{P}_c^+ = P_c^- - L_{c,\text{eff}} \mathbf{U}_c^\top - \mathbf{U}_c L_{c,\text{eff}}^\top + L_{c,\text{eff}} \mathbf{\Gamma}_c L_{c,\text{eff}}^\top. \quad (50)$$

This covariance is not an exact posterior conditional on the realized V_c ; it is the observation-averaged error covariance of the frozen-gain estimator under that local model.

The published physical-frame boundary is

$$\begin{aligned} \mathbf{S}_{c+1}^- &= \mathbf{\Phi}_{c,C_c} \bar{\mathbf{S}}_c^+, \\ \mathbf{P}_{c+1}^- &= \mathbf{\Phi}_{c,C_c} \bar{\mathbf{P}}_c^+ \mathbf{\Phi}_{c,C_c}^\top + \mathbf{\Omega}_c^{\text{chunk}}. \end{aligned} \quad (51)$$

For block-isotropic token noises, the open-loop end-of-chunk aggregate in mode a is

$$v_{c,a}^{\text{chunk}} = \sum_{s=1}^{C_c} v_{c,s,a} \prod_{u=s+1}^{C_c} \rho_{c,u,a}^2. \quad (52)$$

The proposed implementation treats this aggregate as independent boundary noise and adds it at publication. This matches the process-only end covariance. It does not match the exact filtered contribution under (15): early process noise also enters later observations, so subsequent corrections are correlated with the final aggregate. The delayed-noise surrogate therefore approximates residual statistics, gains, and the final filtered MSE.

Proposition 5.1: Causal Chunkwise Implementation

For the co-rotating delayed-noise model (40) and raw frozen gains (41)–(42):

1. the WY system (44)–(45) equals the frozen-gain sequential-residual recurrence;
2. (46) gives every read-after-write prefix without future inputs;
3. (47) exactly represents the committed mean, and (49) is its ensemble-MSE covariance; and
4. (36)–(51) preserve deterministic token-timed transport.

These statements do not assert equivalence to tokenwise Kalman covariance feedback.

Proof. See Appendix B.

5.4. Exact and Approximate Contract

Exact in the proposed implementation	Approximate relative to (17)
Structured prefix damping and rotation. Transported measurement and read addresses.	All raw gains use the same chunk-entry covariance. Measurement contraction does not alter later within-chunk gains.
Causal sequential residuals and read prefixes.	Process noise does not affect within-chunk gains or residual statistics.
Effective-gain mean and Joseph ensemble MSE under the delayed-noise surrogate.	The boundary covariance is neither a realized-value posterior nor the exact filtered MSE under token-local process noise.
Position-dependent tail transport of every write.	Dense Riccati feedback remains delayed to the next chunk.

At $C = 1$ and zero process noise, the proposed implementation reduces to the exact scalar-observation correction in a changed frame. Increasing C exposes more matrix-shaped work but lengthens the interval over which covariance feedback is frozen.

5.5. Complexity and Systems Scope

For state width d_s , value width d_v , and chunk length C , the recurrent core costs

$$O(d_s^2 C + d_s C^2 + d_s C d_v + C^2 d_v + C^3) \quad (53)$$

per chunk, while structured address transport and boundary publication add $O(C d_s + d_s d_v + d_s^2)$. Across T tokens, the recurrent core costs

$$O\left(T d_s^2 + T d_s C + T d_s d_v + T C d_v + T C^2 + \frac{T}{C}(d_s^2 + d_s d_v)\right). \quad (54)$$

In the intended regime $C \leq d_s$ and $d_v \asymp d_s$, covariance–address multiplication has the leading $O(T d_s^2)$ scaling. Chunking changes its execution granularity from dependent matrix–vector products to matrix-shaped work; it does not remove that term or by itself establish a faster layer. These counts cover the recurrent core and exclude input/output projections and other full-layer peripherals.

The implementation primarily targets training and prompt prefill. Autoregressive decode still needs a dense covariance–address product for each generated token and a consistent policy for partial chunks. Complete benchmarks must include the dependent recurrence, Joseph update, rotational maps, state traffic, tail masks, and backward checkpointing. We treat $C = 32$ as an initial engineering point and compare $C \in \{16, 32, 64\}$ by the joint quality–latency–memory tradeoff; component microbenchmarks are not evidence of an end-to-end speedup.

6. Evaluation Protocol

We test KDN-2’s scientific hypothesis separately from the implementation hypothesis. The primary hypothesis is that damped rotational dynamics improve closed-loop associative memory by separating survival from visibility and allowing recall to be controlled by a cue. The secondary hypothesis is that the chunkwise approximation in Section 5 can execute that model with acceptable error and systems cost. We report no KDN-2 result before the corresponding complete implementation is measured; predecessor checkpoints are baselines, not evidence for either claim.

6.1. Experimental Setup

Backbone and pretraining. We use the controlled recurrent-mixer protocol of GDN-2 [Hatamizadeh et al., 2026]. Models are pretrained from scratch on FineWeb-Edu [Penedo et al., 2024] with 4K sequences, a global batch of 0.5M tokens, and identical data seed, optimizer, schedule, surrounding backbone, and non-embedding parameter budget. Feed-forward width is calibrated whenever the mixer changes the recurrent-state parameter count.

Baselines and controls. The main baselines are GDN-2, KDA, isotropic KDN, diagonal KDN, and the SISO and MIMO variants of Mamba-3. The mandatory no-rotation control uses the same paired representation, latent lifts, mode gates, state dimension, and total parameter budget as KDN-2, but fixes $\theta = 0$. A matched additive control keeps KDN-2’s radii, angles, measurement addresses, and read addresses, but replaces the innovation and covariance-shaped gain by an additive outer-product write. This separates the shared rotational geometry from Kalman correction. Additional controls use signed-real poles, damped rotations without a mode gate, and damped rotations with aligned or cross-quadrature routing. Scalar, diagonal, and dense covariance ablations isolate uncertainty structure from process geometry. At small widths, exact tokenwise filtering is the semantic reference for the implementation approximation.

Tasks. The diagnostic suite begins with lag-resolved multi-query associative recall, survival-visibility interventions, randomized-time cue recall, key collisions, and controlled distractors. These tasks make the claimed mechanism identifiable. End-to-end evaluation then uses seeded single- and multi-key RULER tasks [Hsieh et al., 2024], language-modeling perplexity, and the same zero-shot suite as the predecessor mixers. Every example records write time, cue time, requested-item order, and within-chunk position.

6.2. Questions About the Model

Can information survive while remaining temporarily invisible? For one tagged association, we measure both the process-only lineage $\Phi_{t:\tau}\kappa_\tau$ and the closed-loop lineage $\Psi_{t:\tau}\kappa_\tau$. At every lag we report norm, visibility under a fixed read direction, and an intervention difference between streams with and without the tagged write. Holding the read fixed prevents a changing query from producing an apparent delayed peak under positive diagonal decay. KDN-2 succeeds on its defining hypothesis only if closed-loop lineage norm can remain high while visibility is low and then rise later.

Is retrieval cue-controlled or merely scheduled? Cue times and requested-item order are randomized independently of write time. We ablate the mode gate, read phase, and optional input-conditioned phase increment, and report both correct recall and periodic false echoes. A fixed oscillator that predicts a delay but ignores the requested content does not count as cue-based recall.

What are the collision and capacity costs? At fixed lag, we vary the number of simultaneously stored pairs, address similarity, and the number of distractors. We report accuracy, effective state rank, and per-mode write/read energy. This tests whether temporal selectivity merely reallocates a fixed state away from content capacity. Rotation is not credited with extra capacity unless it improves the matched-dimension collision curve.

What does dense uncertainty contribute? At identical mean state, process, and parameter budget, we compare full P with $\text{diag}(P)$ and $\text{Tr}(P)I/d_s$. Results are stratified by key collision and by alignment with prior-covariance eigenvectors. We also measure the angles between $\hat{P}_t h_t$, h_t , and the eventual read direction. This identifies a central risk: the statistically preferred correction can rotate away from the geometry useful for retrieval.

6.3. Questions About the Implementation

How much error does chunkwise inference introduce? At small d_s and T , an FP64 implementation of (17) supplies the reference. We measure output, mean-state, covariance, and gradient error for $C \in \{1, 16, 32, 64\}$ while sweeping address correlation, observation noise, process noise, and rotational rate. The comparison is reported by position within a chunk. This directly exposes gain-freezing error, delayed process-noise error, and any residual boundary bias. The $C = 1$, zero-process-noise case must match the reference under the same read-after-write convention.

Does the implementation preserve the intended token timing? We inject identical writes at every within-chunk offset and compare their end-of-chunk tail transport with the appropriate remaining transition product. Shifting chunk boundaries must not change deterministic rotational phase. Long-horizon, large-angle, repeated-key, partial-tail, and poisoned-boundary tests verify that each chunk consumes the state actually published by its predecessor.

Is the complete layer faster? Correctness is checked against an independent recurrence for forward outputs, published states, covariance symmetry and positivity, and gradients. The primary H200 benchmark measures the full dependent layer at the production model geometry. It includes address/value projections in the shared mixer contract, transported addresses, causal reads, Joseph updates, rotational publication, tail masks, state traffic, and the actual backward checkpoint policy. We report training forward-backward, prefill, and decode separately, together with peak memory and quality. Isolated covariance products or triangular solves are diagnostic microbenchmarks, not a speed result. The detailed protocol appears in Appendix F.

7. Related Work

Linear attention and delta-rule memories. Linear attention replaces the softmax attention matrix with a recurrent fast-weight state [Katharopoulos et al., 2020]. DeltaNet interprets that state as an online associative memory and writes the error at the current key [Schlag et al., 2021; Yang et al., 2024b]. Gated DeltaNet, GDN-2, and KDA add learned forgetting or richer positive real transitions [Yang et al., 2024a; Hatamizadeh et al., 2026; Kimi Team, 2025]. KDN-2 instead gives surviving associations explicit phase motion while retaining uncertainty-aware residual correction.

Complex and rotational dynamics. Complex poles and phase-aware decay appear in recurrent sequence models such as RetNet [Sun et al., 2023]; signed and oscillatory eigenvalues can also improve linear state tracking [Grazzi et al., 2025]. Most closely, Mamba-3 combines accumulated rotary query and key maps with an exponential-trapezoidal SISO or MIMO recurrence [Lahoti et al., 2026]. Section 4.4 shows that its rotary addressing and KDN-2’s rotational state transport are coordinate-equivalent and yield the same open-loop relative-phase kernel. KDN-2 does not claim rotation itself as the distinction; it couples that geometry to innovation-based writes and dense Kalman uncertainty, and studies the resulting closed-loop survival, visibility, and cue control.

Kalman filtering and online least squares. The Kalman filter is the exact recursive estimator for linear-Gaussian state-space models [Grewal, 2025]. Recursive least squares is its static-state specialization and an online second-order update. Recent Kalman sequence mixers use structured information-form recurrences for parallel execution [Shaj et al., 2026]. KDN-2 uses the same filtering principle to couple a rotational associative process with uncertainty-aware writes. Its chunkwise dense-covariance recursion is an implementation approximation, not a different filtering model.

Test-time training and recurrent online learners. TTT layers treat hidden state as the parameters of a local learner and update it on input chunks [Sun et al., 2024]; Titans likewise emphasizes learned long-term neural memory [Behrouz et al., 2024]. The KDN-2 implementation shares the use of matrix-oriented local adaptation, but its preconditioner and error state come from the Kalman model, and its WY system preserves causal residual prefixes.

8. Limitations and Future Directions

Rotation does not create storage capacity. Finite frequency banks can alias; periodic modes can resurface stale content; learned angles can collapse toward zero; and an apparently hidden quadrature can be erased by later measurement updates. A unit-radius mode with persistent process noise may also accumulate unbounded uncertainty when it is poorly observed. Stable radii, matched zero-angle controls, and closed-loop lineage measurements are necessary but do not remove these risks.

A fixed oscillator can solve fixed-delay data without learning arbitrary recall on cue. Query-controlled observability is therefore indispensable to the central claim. Even then, a single phase kick moves every association sharing a mode, so more expressive active/archive routing may trade temporal control against content interference.

Dense uncertainty introduces a separate geometric risk. The Kalman direction $\hat{P}_t h_t$ minimizes estimation error under the stated model, but it may point away from both the measurement address and the later read. Better state estimation need not imply better associative retrieval; gain-read alignment must be measured rather than assumed.

The practical inference path is approximate. Frozen gains are order sensitive, process noise is delayed inside a chunk, and the propagated Joseph matrix is an ensemble MSE rather than the posterior covariance conditional on a realized neural-network history. Learned addresses and noise scales also depart from literal fixed-design assumptions. These approximations may produce chunk-length and within-chunk-position effects even though deterministic rotation is transported exactly.

Finally, KDN-2 retains a quadratic covariance state and $O(Td_s^2)$ covariance-address work. Chunking changes execution granularity, not asymptotic cost, and autoregressive decode remains especially demanding. Future work should first improve cue-controlled read and active/archive routing, then study richer frequency mixtures, process-noise treatment inside chunks, and explicitly ablated covariance compression. Each changes a distinct part of KDN-2 and must be evaluated without conflating model geometry with implementation speed.

9. Conclusion

KDN-2 extends recurrent associative memory beyond geometric forgetting. Its damped rotational process separates how long information survives from how that information moves in phase, while phase-aware query routing determines when surviving content is visible. This rotational geometry is shared with Mamba-3 under an exact change of coordinates; KDN-2’s contribution is to couple it to covariance-shaped residual correction. The survival-visibility distinction is meaningful only in the closed loop: later residual corrections can erase a hidden mode, and a fixed oscillator can produce scheduled echoes without learning recall on cue. KDN-2 therefore couples rotational dynamics to uncertainty-aware associative writes and evaluates survival, visibility, and cue control separately. Exact tokenwise inference supplies the model semantics; the co-rotating dense-covariance recursion is a proposed approximate implementation. The central empirical question is whether this richer memory geometry improves closed-loop recall without unacceptable interference, approximation error, or systems cost.

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A. From Conditional MSE to the Kalman Gain

Condition on $\mathcal{F}_{t-1}$ and the current measurement address $\mathbf{h}_t$; the other current design quantities in (15) are also treated as known. For value coordinate j , let

$$\hat{\mathbf{s}}_{t,j} = \mathbb{E}[\mathbf{s}_{t,j}^* \mid \mathcal{F}_{t-1}, \mathbf{h}_t], \quad \boldsymbol{\epsilon}_{t,j}^- = \mathbf{s}_{t,j}^* - \hat{\mathbf{s}}_{t,j}, \quad \text{Cov}(\boldsymbol{\epsilon}_{t,j}^- \mid \mathcal{F}_{t-1}, \mathbf{h}_t) = \hat{\mathbf{P}}_t. \quad (55)$$

The scalar observation for this value coordinate is

$$v_{t,j} = \mathbf{h}_t^\top \mathbf{s}_{t,j}^* + \zeta_{t,j}, \quad \mathbb{E}[\zeta_{t,j}] = 0, \quad \text{Var}(\zeta_{t,j}) = r_t, \quad (56)$$

where $\zeta_{t,j}$ is independent of the predictive error. Starting from the objective, consider the affine residual family

$$\tilde{\mathbf{S}}_t(\boldsymbol{\kappa}_t) = \hat{\mathbf{S}}_t + \boldsymbol{\kappa}_t(\mathbf{v}_t - \hat{\mathbf{S}}_t^\top \mathbf{h}_t)^\top. \quad (57)$$

Its error in column j is

$$\mathbf{s}_{t,j}^* - \tilde{\mathbf{s}}_{t,j}(\boldsymbol{\kappa}_t) = (\mathbf{I} - \boldsymbol{\kappa}_t \mathbf{h}_t^\top) \boldsymbol{\epsilon}_{t,j}^- - \boldsymbol{\kappa}_t \zeta_{t,j}. \quad (58)$$

The conditional error covariance is therefore

$$\mathbf{P}_t(\boldsymbol{\kappa}_t) = (\mathbf{I} - \boldsymbol{\kappa}_t \mathbf{h}_t^\top) \hat{\mathbf{P}}_t (\mathbf{I} - \boldsymbol{\kappa}_t \mathbf{h}_t^\top)^\top + r_t \boldsymbol{\kappa}_t \boldsymbol{\kappa}_t^\top. \quad (59)$$

Because all d_v value coordinates share this covariance,

$$\begin{aligned} \arg \min_{\boldsymbol{\kappa}_t} \mathbb{E} \left[\|\mathbf{S}_t^* - \tilde{\mathbf{S}}_t(\boldsymbol{\kappa}_t)\|_F^2 \middle| \mathcal{F}_{t-1}, \mathbf{h}_t \right] \\ = \arg \min_{\boldsymbol{\kappa}_t} \text{Tr}(\mathbf{P}_t(\boldsymbol{\kappa}_t)). \end{aligned} \quad (60)$$

Indeed, the first objective equals $d_v \text{Tr}(\mathbf{P}_t(\boldsymbol{\kappa}_t))$. Expanding the trace gives

$$\text{Tr}(\mathbf{P}_t(\boldsymbol{\kappa}_t)) = \text{Tr}(\hat{\mathbf{P}}_t) - 2\boldsymbol{\kappa}_t^\top \hat{\mathbf{P}}_t \mathbf{h}_t + (r_t + \mathbf{h}_t^\top \hat{\mathbf{P}}_t \mathbf{h}_t) \boldsymbol{\kappa}_t^\top \boldsymbol{\kappa}_t. \quad (61)$$

Its Hessian is $2(r_t + \mathbf{h}_t^\top \hat{\mathbf{P}}_t \mathbf{h}_t) \mathbf{I} \succ 0$. Hence the unique minimizer is

$$\boxed{\boldsymbol{\kappa}_t^* = \frac{\hat{\mathbf{P}}_t \mathbf{h}_t}{r_t + \mathbf{h}_t^\top \hat{\mathbf{P}}_t \mathbf{h}_t}}. \quad (62)$$

Thus the optimal Kalman gain minimizes conditional mean-squared error and, equivalently, the sum of post-correction error variances. Under joint Gaussianity the resulting estimator is the conditional mean given $\mathcal{F}_t = \sigma(\mathcal{F}_{t-1}, \mathbf{h}_t, \mathbf{v}_t)$; with only the stated second moments it is the unique linear-MMSE residual estimator.

B. Proofs for the Chunkwise Implementation

All quantities in this section are expressed in the co-rotating entry frame, and we suppress the chunk subscript and tildes for readability.

B.1. WY Equivalence and Causality

Let the explicit frozen-gain recurrence be

$$\begin{aligned} \mathbf{X}_0^{\text{virt}} &= \mathbf{S}^-, \\ \mathbf{e}_t &= \mathbf{v}_t - (\mathbf{X}_{t-1}^{\text{virt}})^\top \mathbf{h}_t, \\ \mathbf{X}_t^{\text{virt}} &= \mathbf{X}_{t-1}^{\text{virt}} + \boldsymbol{\ell}_t \mathbf{e}_t^\top. \end{aligned} \quad (63)$$

Expanding its prefix gives

$$\mathbf{X}_{t-1}^{\text{virt}} = \mathbf{S}^- + \sum_{s < t} \boldsymbol{\ell}_s \mathbf{e}_s^\top. \quad (64)$$

Substitution into the innovation yields

$$\mathbf{e}_t^\top = \Delta_t - \sum_{s < t} (\mathbf{h}_t^\top \ell_s) \mathbf{e}_s^\top. \quad (65)$$

Stacking rows gives

$$[\mathbf{I} + \text{tril}(\mathbf{H}^\top \mathbf{L}, -1)] \mathbf{Z} = \Delta. \quad (66)$$

The bracket is T from (44); therefore $\mathbf{Z} = \bar{\mathbf{A}} \Delta$ and

$$\mathbf{X}_C^{\text{virt}} = \mathbf{S}^- + \mathbf{LZ} = \bar{\mathbf{S}}^+. \quad (67)$$

At output position j , the virtual state contains exactly writes $1:j$, so

$$\mathbf{o}_j = (\mathbf{S}^-)^\top \mathbf{g}_j + \sum_{t \leq j} (\mathbf{g}_j^\top \ell_t) \mathbf{e}_t. \quad (68)$$

Stacking these rows gives (46). Equivalently, every leading principal block of the unit-lower matrix T is independent of later rows, so neither $\mathbf{Z}_{1:j}$ nor output j can use a future token.

B.2. Effective Gain and Joseph Ensemble MSE

The mean identity follows from

$$\bar{\mathbf{S}}^+ = \mathbf{S}^- + \mathbf{L} \bar{\mathbf{A}} \Delta = \mathbf{S}^- + \mathbf{L}_{\text{eff}} \Delta. \quad (69)$$

For one value coordinate, let $\epsilon^- = \mathbf{x}^* - \mathbf{s}^-$. Under (40),

$$\Delta_{:,j} = \mathbf{H}^\top \epsilon^- + \xi_{:,j}. \quad (70)$$

The committed error is

$$\epsilon^+ = (\mathbf{I} - \mathbf{L}_{\text{eff}} \mathbf{H}^\top) \epsilon^- - \mathbf{L}_{\text{eff}} \xi. \quad (71)$$

Taking the second moment and using independence gives

$$\mathbb{E}[\epsilon^+ \epsilon^{+\top}] = \mathbf{A} \mathbf{P}^- \mathbf{A}^\top + \mathbf{L}_{\text{eff}} \mathbf{R} \mathbf{L}_{\text{eff}}^\top, \quad (72)$$

which is (49). Expanding $\mathbf{A}$ and using $\mathbf{U} = \mathbf{P}^- \mathbf{H}$ and $\mathbf{\Gamma} = \mathbf{R} + \mathbf{H}^\top \mathbf{P}^- \mathbf{H}$ gives (50). For all value coordinates,

$$\mathbb{E}[\|\mathbf{X}^* - \bar{\mathbf{S}}^+\|_F^2] = d_v \text{Tr}(\bar{\mathbf{P}}^+). \quad (73)$$

This equality calibrates the covariance of the chosen estimator; it does not claim that its frozen gains minimize the joint chunk objective.

B.3. Token-Timed Transport

The entry-frame correction produced at token t is $\ell_t \mathbf{e}_t^\top$. Its physical direction at that token is $\Phi_{c,t} \ell_t$. At a later token $j \geq t$, deterministic tail transport gives

$$\mathbf{D}_{c,j} \cdots \mathbf{D}_{c,t+1} \Phi_{c,t} \ell_t = \Phi_{c,j} \ell_t. \quad (74)$$

Its read coefficient is

$$\mathbf{g}_{c,j}^\top \Phi_{c,j} \ell_t = \tilde{\mathbf{g}}_{c,j}^\top \ell_t, \quad (75)$$

which is exactly the coefficient in (46). At the boundary, setting $j = C_c$ gives $\Phi_{c,C_c} \ell_t$, so earlier and later writes receive their correct remaining transition products. This proves deterministic token-timed transport without forming an inverse prefix.

For independent block-isotropic process noises, noise injected after micro-step s is transported through steps $s + 1:C_c$. Rotations cancel in its covariance and the remaining scale is $\prod_{u=s+1}^{C_c} \rho_{c,u,a}^2$. Summing independent contributions proves (52). Adding that aggregate only at publication reproduces its open-loop process-only end covariance. It does not reproduce the filtered MSE under the original tokenwise model, because early noise enters later observations and becomes correlated with their corrections. More explicitly, stack the micro-noises for one value coordinate in w , and let B map them to the within-chunk observation vector while G maps them to the final physical state. Their contribution to the true boundary error is

$$(G - \Phi_{c,C_c} L_{c,\text{eff}} B) w, \quad (76)$$

whose covariance contains the correction term. The publication-only surrogate instead uses

$$G \text{Cov}(w) G^\top = \Omega_c^{\text{chunk}}. \quad (77)$$

C. Approximation Oracles

C.1. Exact Joint Conditioning in the Co-Rotating Frame

For a Gaussian entry state and the local observation model (40), exact joint conditioning would use

$$L_{\text{block}} = U \Gamma^{-1}, \quad (78)$$

and

$$\begin{aligned} S_{\text{block}}^+ &= S^- + L_{\text{block}} \Delta, \\ P_{\text{block}}^+ &= P^- - U \Gamma^{-1} U^\top. \end{aligned} \quad (79)$$

The implemented effective gain is instead

$$L_{\text{eff}} = U \Lambda^{-1} \left[I + \text{tril}(H^\top U \Lambda^{-1}, -1) \right]^{-1}. \quad (80)$$

They coincide for $C = 1$ and when Γ is diagonal in the selected token order. In general, exact joint conditioning uses the symmetric innovation inverse, while the implementation uses marginal raw gains and causal lower-triangular residual feedback. Repeated or collinear addresses and small observation noise are therefore mandatory stress cases.

C.2. Repeated-Key Oracle

For one scalar entry state with prior variance p , C identical unit addresses, and observation noise r , the frozen gain and error contraction are

$$\ell = \frac{p}{p+r}, \quad a = 1 - \ell = \frac{r}{p+r}. \quad (81)$$

The error recurrence is $\epsilon_t = a \epsilon_{t-1} - \ell \zeta_t$. Hence

$$\epsilon_C = a^C \epsilon_0 - \ell \sum_{t=1}^C a^{C-t} \zeta_t, \quad (82)$$

and

$$\mathbb{E}[\epsilon_C^2] = a^{2C}p + \ell^2r \frac{1 - a^{2C}}{1 - a^2}. \quad (83)$$

The MSE stays bounded for $0 < a < 1$, but it differs from recursive Kalman filtering because later gains do not shrink. Equation (83) is both a numerical oracle and a direct measure of gain-freezing error.

D. Proofs for the Rotational Process

D.1. Composition and Norm Survival

Proof of Proposition 4.1. Two planar rotations satisfy

$$R(\theta_2)R(\theta_1) = R(\theta_1 + \theta_2). \quad (84)$$

Scalar radii commute with the rotations, and the Kronecker identity acts independently on every content coordinate. Repeated multiplication gives (22). Since $R(\theta)$ is orthogonal,

$$\|[R(\theta) \otimes I_{r_a}]\mathbf{x}\|_2 = \|\mathbf{x}\|_2, \quad (85)$$

so only the product of radii changes the norm, proving (23). $\square$

D.2. Observability of a Hidden Quadrature

The determinant of the matrix in (34) is $-\rho \sin \theta$. It is therefore full rank exactly when $\rho \sin \theta \neq 0$. Small ρ destroys the signal before retrieval, while small $|\sin \theta|$ makes recovery poorly conditioned. This proves only open-loop read observability; intervening correction factors in $\Psi_{t:\tau}$ can still alter or erase the hidden direction.

E. Reference Algorithm and Validation Contract

Initialization. For each batch-head item, initialize

$$S_0^- = \mathbf{0}, \quad P_0^- = \mu^{-1}I, \quad \mu > 0. \quad (86)$$

Reference chunk step. For every real chunk:

1. form every structured prefix $\Phi_{c,t}$ and the transported addresses $\tilde{H}_c, \tilde{Q}_c$;
2. form U_c, Γ_c, Λ_c , and L_c from (41);
3. form Δ_c and solve the unit-lower system for Z_c ;
4. compute causal outputs and commit $\bar{S}_c^+$;
5. form $L_{c,\text{eff}}$ and compute $\bar{P}_c^+$ in Joseph form; and
6. publish (S_{c+1}^-, P_{c+1}^-) with (51).

Required identities. An independent FP64 implementation checks

$$\begin{aligned} Z_{t,:} &= \mathbf{e}_t^\top && \text{from (63),} \\ S^- + LZ &= S^- + L_{\text{eff}}\Delta, \\ \bar{P}_{\text{Joseph}}^+ &= \bar{P}_{\text{expanded}}^+, \\ \bar{P}^+ &= \bar{P}^{+\top}, \quad \lambda_{\min}(\bar{P}^+) > 0 \quad \text{within numerical tolerance.} \end{aligned} \quad (87)$$

Monte Carlo simulation of (40) must match the Joseph matrix to the empirical ensemble MSE. Separately, deterministic writes at every chunk offset must satisfy (74).

Gradient and numerical validation. Gradients with respect to transported addresses, values, noise scales, initial covariance, radii, angles, and latent lifts are compared with an autodifferentiated reference. Tests cover non-multiple tail chunks, low observation noise, repeated addresses, large rotations, and losses on both outputs and final states. The validation oracle keeps covariance operations in FP64, uses Joseph form, symmetrizes only for validation eigenspectrum checks, and masks padded positions from gains, triangular systems, reads, and transition products.

F. Complete Systems Protocol

The primary layer benchmark uses the production geometry $B = 4$, $T = 4096$, $d_{\text{model}} = 2048$, 16 heads, and $d_s = d_v = 128$ on an H200. Every timed call includes the shared projection contract, the dependent chunk recurrence, transported addresses, causal reads, tail masks, state publication, symmetry policy, Joseph update, rotational mean and covariance maps, output normalization, and the actual backward checkpoint policy. Complete forward and complete forward-backward are distinct measurements.

The production covariance, denominators, WY system, and Joseph update use IEEE FP32 after parity against the FP64 validation oracle. A production path may use BF16 for peripheral tensors while keeping recurrent state in FP32. TF32 or lower-precision covariance paths are separate ablations and must first match full-sequence outputs, states, eigenvalue drift, and gradients.

Comparisons run in one process and stream with source revisions and compiled kernels pinned. After 15 warmups, we collect 40 samples in balanced interleaved AB/BA order. We report median and minimum latency, tokens per second, allocated and device-level peak memory, and kernel resource data including registers, shared memory, spills, scratch, occupancy, and synchronization. No layer result is assembled from independently best component timings.

Training, prefill, and decode are reported separately. Decode results include ordinary-token latency, boundary-tail latency, time to first token, and cache bytes. Chunk lengths are selected on a development workload only, with recurrent width, parameter count, precision, batch, sequence length, and training tokens matched. Results are presented as a quality-latency-memory tradeoff; until this complete protocol is run, the chunkwise path is an implementation hypothesis rather than a demonstrated speedup.